Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Predictive Modeling, Time-Series Forecasting (SARIMA), Autonomous Machine Learning, Anomaly Detection, Agentic AI & Prompt Orchestration, Generative AI & LLMs, Capacity Planning & Workforce Optimization, Distributed Data Pipelines, Dimensional Modeling, Automated Testing & QA, Data Governance & Lineage.

Programming Languages & Tools: Python, SQL, R, XGBoost, Scikit-Learn, Random Forests, SARIMA, AWS (S3, Redshift, SageMaker, Lambda, Glue), Azure (ADF, Databricks, Synapse, Azure SQL), Snowflake, Apache Spark, PySpark, Power BI, Tableau, Collibra, KIRO IDE, GitHub Copilot.

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data Scientist & AI Solutions Lead — **May 2022 – Present**

- **Architected autonomous agentic workflows and prompt orchestration** utilizing custom generative AI tools (KIRO IDE, GitHub Copilot) to automate schema discovery, data profiling, and code migration, reducing manual delivery turnaround by 40-50%.
- **Engineered autonomous machine learning anomaly detection frameworks** identifying distribution drift, schema variations, and ETL discrepancies with 95% precision to safeguard downstream BI and pipeline integrity.
- **Built and deployed predictive machine learning classification models** leveraging XGBoost, Random Forests, and Scikit-Learn to forecast customer behavior and incident trends, delivering actionable signals that improved operational reliability by 20%.
- **Designed advanced time-series forecasting models using SARIMA** to predict field support demand and resource capacity requirements, optimizing workforce allocation and reducing operational response times by 20%.
- **Engineered automated testing and evaluation frameworks for Conversational AI** platforms (SpectrumIQ), cutting manual QA turnaround by 50% and accelerating production release cycles for enterprise virtual assistants.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Architected high-throughput distributed data pipelines and dimensional models** (Star/Snowflake schemas, SCD Type 1/2) using Azure (ADF, Databricks, Synapse) and AWS (S3, Redshift) to process hundreds of terabytes with 99.9% availability.
- **Developed automated source-to-target reconciliation firewalls and data lineage** using Collibra to enforce strict data governance and reduce operational incident resolution turnaround by over 80%.
- **Translated complex transactional telemetry into executive corporate strategy** by building Key Operating Metric (KOM) dashboards and semantic models in Power BI (DAX) and Tableau.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed scalable data solutions and analytical intelligence frameworks** supporting global infrastructure and operational excellence initiatives.
- **Partnered with cross-functional engineering teams** to productionize machine learning models and automated monitoring frameworks ensuring operational data integrity.
- **Communicated complex analytical insights and model performance metrics** to senior leadership through executive presentations and detailed technical documentation.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.